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# Lab 10.1 Machine Learning 3 - Real-time Sound Recognition for Classification

## Utilizing MTConnect sound stream

## Learning goals

Students will be able to:

1. Understand and utilize MTConnect sound data stream
2. Implement a CNN model to predict operational state of the vacuump pump
3. Build a real-time web-based monitoring system

## 1.1 Introduction

In this lab, we will implement real-time machine sound recognition using a CNN model and MTConnect sound stream to predict the operational state of the vacuum pump. This lab is broken down into three main sections: **1) Utilize MTConnect sound data stream**, **2) Implement a CNN model you developed in the prelab using TensorFlow Lite**, and **3) Build up the entire monitoring system.**

Note that all sample codes (almost complete) for this lab are available on [Github repo](). If you want, please download them to Raspberry Pi.

To get started, Figure 1 illustrates a simplified sound recognition system. IIoT technologies and embedded machine learning models realize this system. The idea is that the Raspberry Pi listens to sound from the machine and recognize the operational state by AI model, and then we are able to monitor remotely the real-time state via a web-based dashboard.

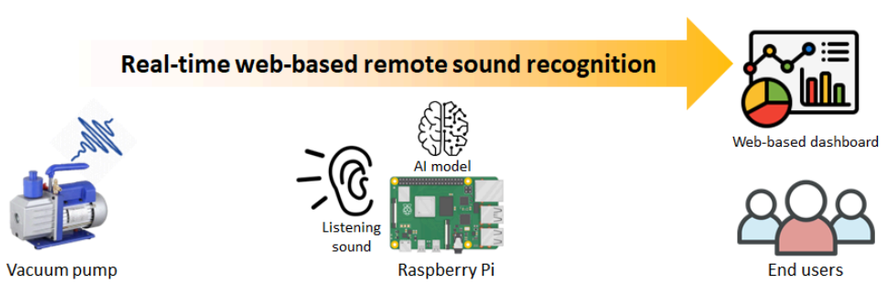


*Figure 1 Simplified system configuration of vacuum pump sound recognition for Lab 10*

Details of the system configuration and data flow are shown in Figure 2. Please take a close look at Figure 2. Let's say the MTConnect stream for sound is "sub-MTConnect" for sound to distinguish it from the MTConnect for results. First, the internal sound sensor connected to the headless computer captures analog sound signals and then converts them to digital signals. Second, the sub-MTConnect adapter samples, formats, and converts them to the sub-MTConnect agent. Therefore, as long as a client is accessible via TCP/IP to the headless computer, it can read the sound stream using the sub-MTConnect agent in real-time. Third, the edge computer (Raspberry Pi) gets the sound stream and takes a signal processing for the input feature of the CNN model

that we developed in Prelab 10. For AI implementation, we utilize TensorFlow Lite on Raspberry Pi. Fourth, the outputs of the CNN model with the AI algorithm are taken to MTConnect. Fifth, the MTConnect data collector program aggregates MTConnect agent data and sends them to the MySQL database. Lastly, Grafana dashboard with the MySQL database can let end users know the current/history running state of the vacuum pump.

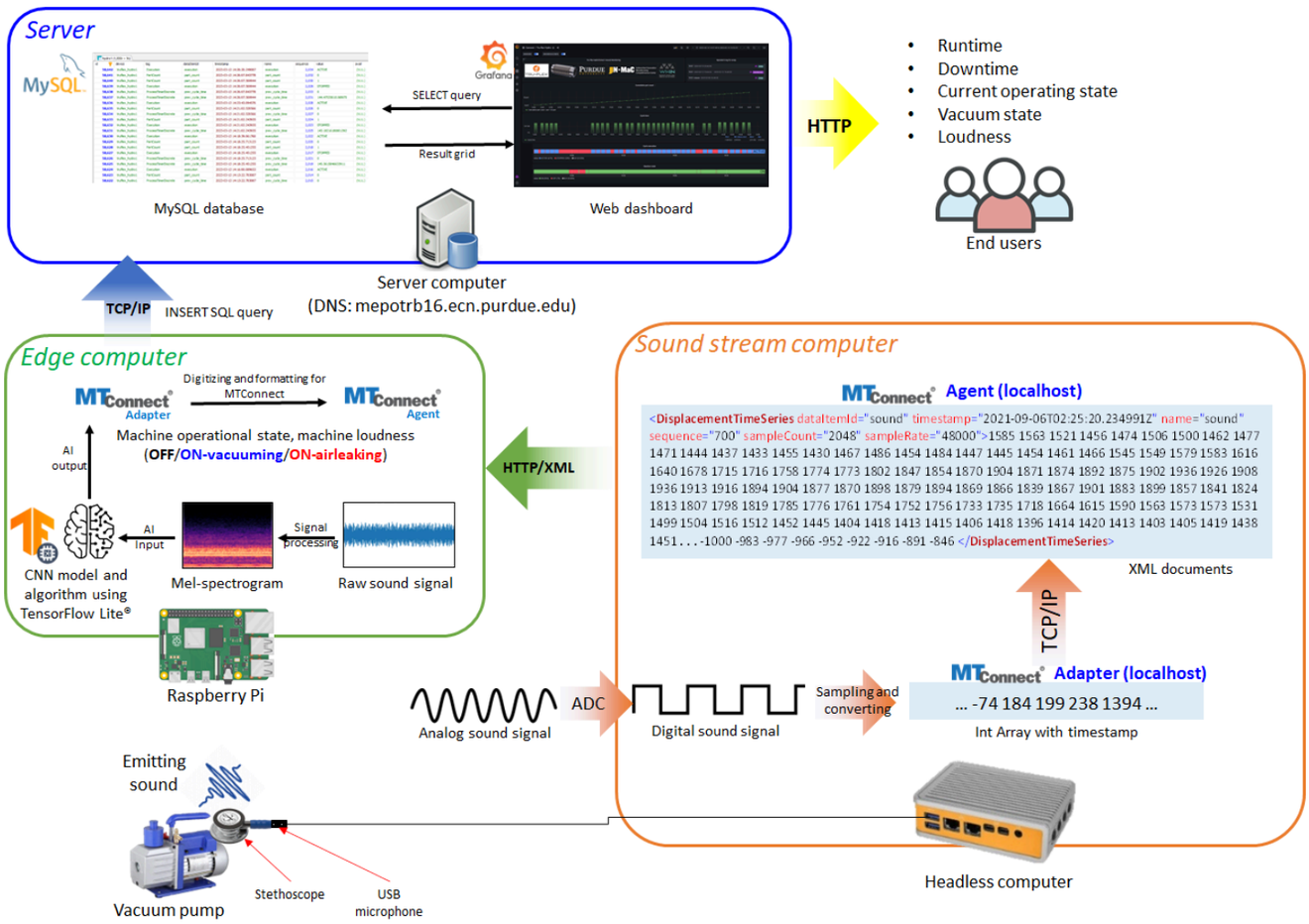


*Figure 2 Detailed system configuration and data flow for Lab 10*

## 1.2 Sensor and server information

Here, sensor specifications and server information (for MySQL and Grafana services) are summarized for your convenience, even though you are able to find the information in the previous prelab and lab manuals.

Sensor specifications including the sub-MTconnect sound stream are in Table 1. Sound pressure level (dB SPL) is calculated by the sensitivity, SNR, gain of PCM (pulse-code modulation) in ALSA (advanced Linux sound architecture), and reference pressure ($2\times10^{-5}$ Pa). For the details of the conversion, please go to reference 1, and reference 2. The gain of PCM is +28.78 dB. Using the information of the sensor and sound stream, perform Task 1.1 to get access to the MTConnect sound stream.

*Table 1 Sound sensor and stream specifications*

| Target | Specification | Value | Unit/remark |
|---|---|---|---|
| USB Microphone | Sensitivity | -38±3 | dB |
| | Output impedance | 680 | Ω |
| | Directivity | Unidirectional | - |
| | Channel | Mono | - |
| | Frequency range | 20 - 24000 | Hz |
| | Operation voltage | 5 | V |
| | Current consumption | 0.5 | mA |
| | SNR (Signal to Noise Ratio) | 56 | dB |

| Target | Specification | Value | Unit/remark |
|---|---|---|---|
| Sound stream | Middleware | MTConnect | |
| | Agent IP address | 192.168.1.4 | Router network (DNS: mepotrb16.ecn.purdue.edu) |
| | Port | 5001 | |
| | Data type | DiscreteTimeseries | |
| | Data format | Signed 16-int | Full scale is -1 to 1. |
| | Sampling frequency | 48 | kHz |
| | Count/sequence | 2048 | Approximately 43 msec |
| | dataitemId | sensor1 | |

## Task 1.1

Using curl command below on Raspberry Pi and laptop, read current sound stream of the sensor and then attach it to the report as follows. Check the timestamp indicates current. It must be working on your laptop as well as long as both computers are on the same router network. Perform the same in the command prompt of your laptop as Raspberry Pi.

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### Raspberry Pi - Terminal AND Windows - Command Prompt

`curl 192.168.1.4:5001/current`

OR

`curl mepotrb16.ecn.purdue.edu:5001/current`

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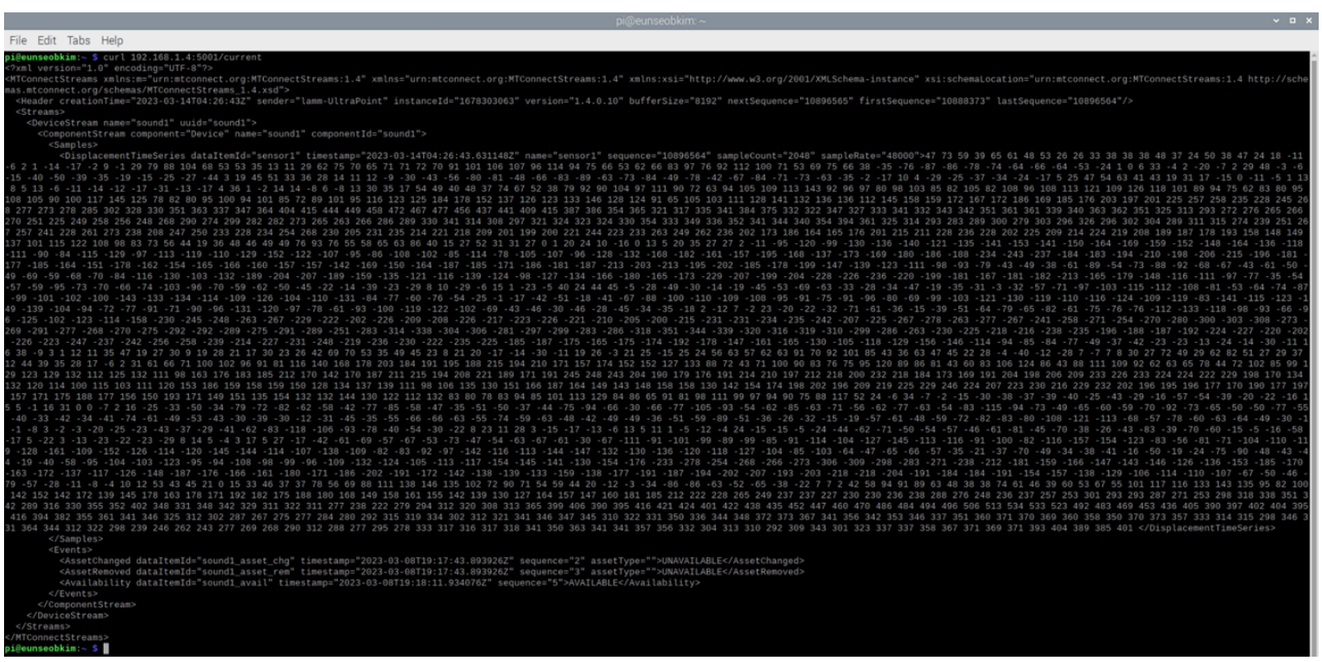
- Note that the headless computer is connected to the router network (DNS: mepotrb16,ecn.purdue.edu). Because of the port forwarding of 5001 to the headless computer (192.168.1.4), we can has the same results if you get access both IP address and DNS.

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192.168.1.23 (WayVNC) - RealVNC Viewer

prelab10 | pi@AaryaFarheen: ~... | prelab10_load_mod... | pi@AaryaFarheen: ~

pi@AaryaFarhe

File Edit Tabs Help

```
(base) pi@AaryaFarheen:~/prelab10 $ curl 192.168.1.4:5001/current
<?xml version="1.0" encoding="UTF-8"?>
<MTConnectStreams xmlns:m="urn:mtconnect.org:MTConnectStreams:1.4" xmlns="urn:mtconnect.org:MTConnectStreams:1.4" xmlns:
mas.mtconnect.org/schemas/MTConnectStreams_1.4.xsd">
  <Header creationTime="2026-04-08T17:30:35Z" sender="lamm-UltraPoint" instanceId="1775668545" version="1.4.0.10" buffer
  <Streams>
    <DeviceStream name="sound1" uuid="sound1">
      <ComponentStream component="Device" name="sound1" componentId="sound1">
        <Samples>
          <DisplacementTimeSeries dataItemId="sensor1" timestamp="2026-04-08T17:30:35.119924Z" name="sensor1" sequence="
9 1082 829 599 409 250 74 -65 -200 -302 -375 -431 -488 -522 -539 -532 -529 -503 -484 -446 -356 -236 -70 113 347 597 868
3 4102 3893 3672 3439 3219 3006 2799 2575 2344 2130 1940 1751 1565 1398 1253 1092 974 881 782 734 711 719 747 782 854 91
51 3653 3653 3663 3631 3595 3560 3494 3372 3261 3134 2979 2846 2702 2501 2318 2114 1946 1790 1662 1542 1483 1405 1320 12
64 609 498 452 388 301 228 156 83 31 0 -39 -41 -16 1 21 -6 -35 -58 -85 -141 -198 -252 -304 -330 -358 -391 -425 -457 -471
7 -51 -19 83 177 206 240 275 310 347 390 431 471 556 653 731 835 946 1108 1233 1360 1446 1539 1658 1767 1857 1958 2078 2
35 621 512 455 365 287 273 259 219 202 205 203 196 194 226 272 329 371 446 539 660 771 912 1049 1199 1391 1593 1787 2005
39 1664 1510 1367 1229 1097 965 888 793 704 640 564 491 426 358 295 273 249 205 186 134 98 51 47 71 69 137 212 337 495 6
297 6209 6053 5933 5784 5638 5466 5259 5093 4916 4754 4523 4299 4087 3864 3666 3474 3282 3132 2967 2802 2642 2522 2369 2
 5514 5825 6081 6282 6423 6541 6647 6673 6680 6653 6590 6518 6426 6294 6112 5924 5728 5497 5235 4995 4738 4476 4198 3889
3 467 567 634 711 800 879 940 997 1039 1043 1044 1030 1048 1031 1054 1103 1137 1168 1203 1215 1224 1209 1183 1157 1057 9
2301 -2296 -2235 -2165 -2125 -2061 -2009 -1942 -1871 -1788 -1703 -1633 -1641 -1631 -1604 -1617 -1637 -1678 -1739 -1824 -
 -4156 -4249 -4317 -4402 -4486 -4570 -4644 -4670 -4700 -4729 -4714 -4697 -4688 -4658 -4598 -4530 -4473 -4409 -4347 -4302
62 -4774 -4803 -4850 -4899 -4968 -5061 -5174 -5292 -5368 -5444 -5531 -5599 -5652 -5632 -5689 -5677 -5643 -5595 -5547 -54
4796 -4859 -4898 -4918 -4949 -4984 -5037 -5113 -5181 -5251 -5345 -5424 -5503 -5594 -5672 -5739 -5817 -5877 -5917 -5956 -
 -4001 -3829 -3650 -3538 -3409 -3273 -3187 -3078 -2967 -2898 -2834 -2747 -2688 -2602 -2598 -2613 -2659 -2740 -2849 -2972
18 -6490 -6462 -6427 -6345 -6243 -6114 -5932 -5716 -5480 -5253 -4983 -4637 -4275 -3937 -3581 -3213 -2817 -2421 -2005 -16
14 -2728 -3052 -3343 -3612 -3868 -4092 -4301 -4436 -4532 -4615 -4669 -4716 -4707 -4722 -4661 -4572 -4497 -4391 -4219 -40
5 2390 2528 2576 2619 2630 2604 2549 2485 2430 2327 2214 2082 1950 1775 1600 1429 1227 1014 816 632 427 185 3 -222 -398
686 1020 1369 1734 2061 2365 2677 2934 3159 3394 3533 3678 3788 3852 3887 3877 3879 3833 3770 3688 3611 3509 3371 3246 3
0 1037 1101 1168 1276 1410 1521 1672 1812 1941 2084 2246 2379 2492 2615 2719 2853 2957 3024 3088 3140 3175 3205 3221 321
8 -392 -424 -468 -482 -498 -504 -452 -391 -301 -196 -70 56 196 347 487 640 785 949 1113 1269 1445 1575 1681 1820 1947 20
234 139 57 0 -60 -88 -123 -106 -110 -68 -32 22 115 214 322 479 666 822 965 1121 1314 1522 1683 1851 2020 2218 2403 2555
0 3274 3148 3020 2936 2839 2727 2637 2551 2446 2369 2273 2192 2172 2122 2067 2043 2042 2103 2124 2172 2228 2293 2371 242
201 3355 3443 3523 3535 3572 3635 3642 3616 3574 3552 3512 3447 3373 3322 3288 3259 3250 3223 3213 3217 3196 3187 3180 3
 1067 961 891 839 813 794 818 865 919 989 1049 1126 1203 1265 1363 1487 1604 1727 1878 2037 2165 2339 2482 2650 2824 298
321 2003 1706 1419 1123 834 552 296 17 -229 -461 -683 -920 -1139 -1329 -1492 -1659 -1795 -1857 -1905 -1952 -1994 -1983 -
0 644 582 465 313 182 35 -112 -266 -470 -678 -904 -1113 -1304 -1470 -1630 -1780 -1912 -2024 -2136 -2259 -2366 -2513 -267
354 -3233 -3108 -2983 -2825 -2683 -2561 -2431 -2273 -2098 -1964 -1810 -1688 -1603 -1536 -1455 -1380 -1378 -1390 -1408 -1
-3288 -3279 -3259 -3220 -3156 -3103 -3050 -2984 -2895 -2806 -2695 -2569 -2467 -2370 -2280 -2243 -2214 -2198 -2170 -2169
6 -3564 -3640 -3735 -3824 -3879 -3932 -3978 -4019 -4042 -4086 -4120 -4147 -4164 -4184 -4205 -4222 -4240 -4239 -4226 -421
762 -2740 -2709 -2710 -2726 -2770 -2836 -2898 -2960 -3032 -3090 -3173 -3264 -3380 -3446 -3496 -3557 -3620 -3677 -3742 -3
-3024 -2938 -2876 -2826 -2789 -2751 -2753 -2770 -2784 -2798 -2891 -2988 -3104 -3289 -3451 -3593 -3765 -3934 -4073 -4227
3 -2125 -1970 -1817 -1718 -1597 -1468 -1320 -1195 -1038 -864 -713 -506 -322 -123 78 227 360 478 606 691 746 774 775 735
-617 -503 -292 -99 85 267 435 634 830 1019 1218 1396 1584 1739 1869 2058 2200 2324 2447 2544 2643 2760 2832 2901 2967 29
-263 -530 -760 -997 -1239 -1443 -1609 -1791 -1939 -2083 -2174 -2269 -2377 -2455 -2506 -2531 -2512 -2530 -2500 -2477 -240
6 627 710 774 818 855 904 926 943 955 933 892 848 771 674 555 418 266 112 -37 -230 -433 -644 -863 -1081 -1259 -1456 -163
3 -344 -71 212 525 778 980 1143 1278 1394 1478 1539 1619 1650 1667 1692 1683 1632 1587 1536 1489 1450 1347 1258 1171 107
 -1384 -1345 -1277 -1205 -1142 -1074 -965 -871 -816 -717 -624 -522 -466 -388 -316 -241 -150 -123 -65 -6 19 66 116 128 14
2274 -2289 -2292 -2322 -2324 -2356 -2315 -2272 -2273 -2243 -2193 -2155 -2115 -2040 -2041 -2054 -2004 -1984 -1996 -2009 -
 -2713 -2781 -2806 -2836 -2884 -2912 -2938 -2929 -2925 -2924 -2923 -2918 -2866 -2863 -2838 -2801 -2731 -2688 -2597 -2520
22 -3469 -3440 -3391 -3340 -3257 -3151 -3053 -2897 -2711 -2543 -2362 -2178 -1998 -1790 -1652 -1505 -1382 -1244 -1100 -94
913 -1140 -1343 -1574 -1819 -2036 -2239 -2449 -2615 -2810 -2918 -3000 -3043 -3066 -3089 -3074 -3019 -2926 -2810 -2641 -2
19 2049 2110 2164 2181 2199 2190 2185 2150 2081 2015 1925 1830 1711 1591 1419 1221 996 791 562 329 80 -140 -401 -653 -88
195 -971 -729 -453 -170 114 441 779 1101 1420 1751 2052 2343 2606 2862 3048 3191 3310 3368 3387 3348 3298 3242 3163 3036
-782 -981 -1194 -1318 -1417 -1423 -1370 </DisplacementTimeSeries>
        </Samples>
        <Events>
          <AssetChanged dataItemId="sound1_asset_chg" timestamp="2026-04-08T17:15:45.950385Z" sequence="2" assetType="">
```

---

## 1.3 Handling MTConnect sound stream in Python

Let's get access to the MTConnect sound stream and handle it using Python. Use lab10_sample1.py which is almost complete other than some constant variables. In the Python script, We will read approximately 1-second-long data at the moment for the CNN model we developed in the prelab. And then, signal processing and sound pressure level calculation will be made. Eventually, we will check the sound signal by plotting them.

The incomplete part of the sample script is below. You have to change the variables which is wrarpped by aserteric mark (*). By completing the sample code , perform Task 1.2.

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**Python - Python3 (Part of lab10_sample1.py)**

```
## == CONSTANT ==
SAMPLE = "sample" # sample string for MTConnect HTML sample method
CURRENT = "current" # current string for MTConnect HTML current method
*SAMP_RATE* = int(?) # sampling rate
*CHUNK* = int(?) # chunk size
*AGENT* = "http://?ip?:?port?/" # MTConnect agent ip:port

*N* = int(?) # number of sequence to take sound
*N_FFT* = int(?) # number of FFT
*N_MELS* = int(?) # number of Mel filter bank
```

---

## Task 1.2

To complete lab10_sample1.py, the required information is below. Note that you can use either Raspberry Pi or your laptop for this task.

- MTConnect (sound stream) agent IP: 192.168.1.4
- MTConnect (sound stream) agent port number: 5001
- Sampling frequency (sampling rate): MTConnect sound stream specification

- Chunk size: MTConnect sound stream specification
- Number of the required sequence: Same as CNN model training to make the sound signal approximately 1-second-long
- Number of FFT data points: Same as the CNN model training
- Number of Mel filter-bank: Same as the CNN model training

1. After completing the sample script, plot the signal as Figure 3 and then attach it on the report.
2. Calculate and print out the exact signal length in the second unit as Figure 4 and then attach the screenshot. You can use the commented-out line (line 57), but you have to replace "?" to get the answer.

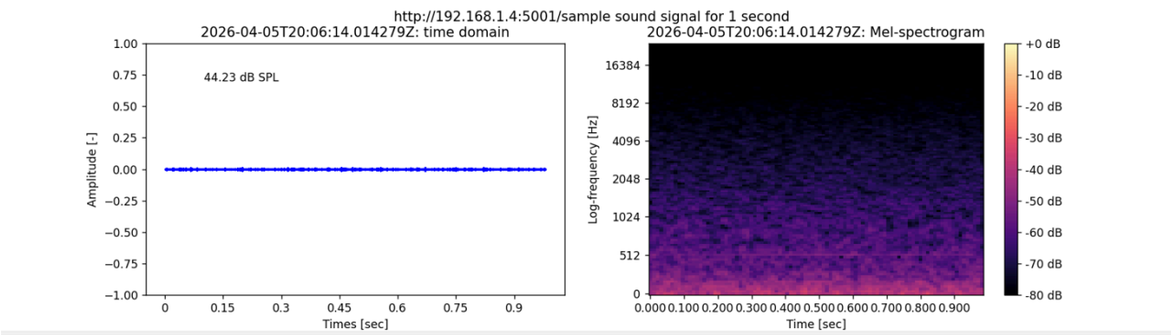


*Figure 3 Sound signal plot: time domain (left) and Mel-spectrogram (right)*

```
Current sequence info, timestamp: 2026-04-05T20:06:14.014279Z, firstSeq=19094, nextSeq=10902, lastSeq=19093
signal SPL (sound pressure level) is 44.23241963823554 dB.
Length of signal is  0.9813333333333333 [sec]
PS C:\Users\nobin>
```
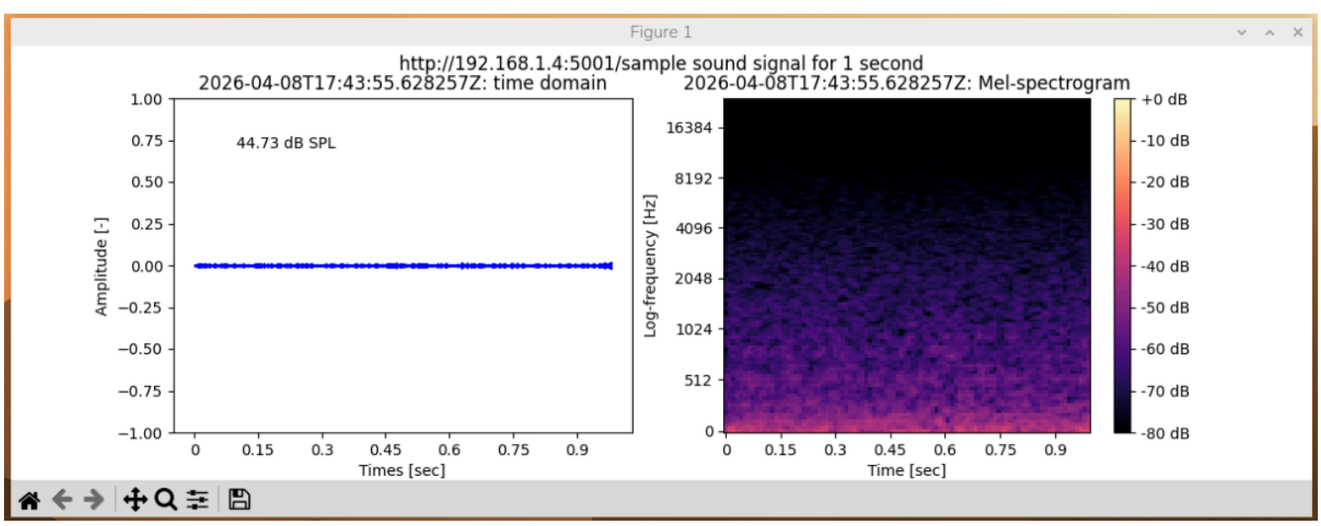
*Figure 4 Terminal after running lab10_sample1.py*